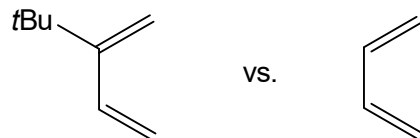
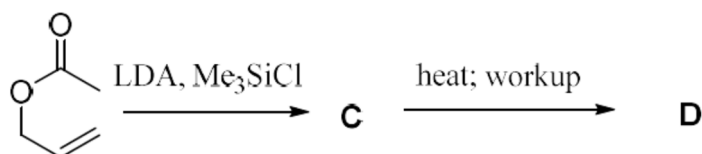


1. Specify and explain the selectivity rules for linear π systems containing 6 electrons and 6 p orbitals, i.e., conrotatory or disrotatory under heat or light conditions.

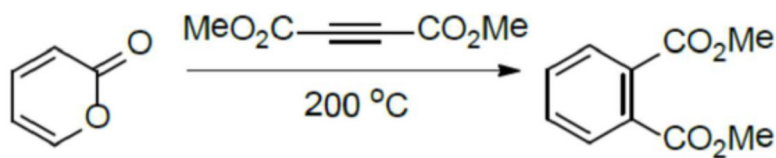
2. Why does 2-*t*Bu-1,3-butadiene undergo much faster Diels-Alder reaction than 1,3-butadiene?



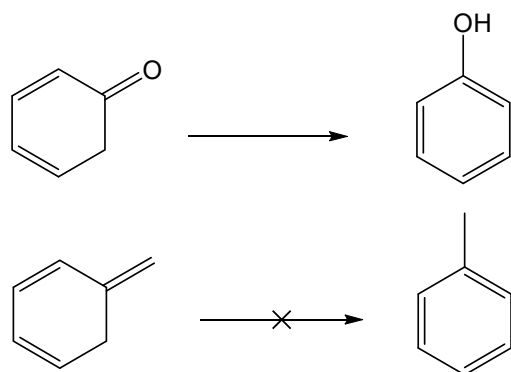
3. Draw the products of the following transformations and explain your answer.



4. Provide a mechanism for the following transformation.



5. Explain why for the following reactions, the first one is easy while the second one is hard.



6. Provide a mechanism for the following transformation.

